STAT 527 Advanced Regression Analysis
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Transforming the predictors

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Transforming the Predictors

Before:

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$$

Now:

$$y = \beta_0 + \beta_1 f_1(x) + \cdots + \beta_q f_q(x) + \epsilon$$

$f_j(x)$ are called basis functions.

Examples:

- Polynomials
- Regression splines

Polynomial Regression

We have seen that when $\mathrm{E}[Y \mid X]$ and X are related by a smooth curve, it is often possible to find a transformation to linearity.

An alternative is to expand the linear regression model to a polynomial regression model. Here is a polynomial regression model of degree d:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_d X^d + \varepsilon$$

When is this design matrix full rank?

The errors ε_i are independent with mean 0 and either constant variance σ^2 or $\mathrm{var}(\varepsilon_i) = \sigma^2 / w_i$ for known w_i.

In this model $\mathrm{E}[Y \mid X]$ is just a linear combination of powers of X, and least squares estimation can still be used.

How to know if polynomial terms should be included

- We add polynomial terms to the model and treat them as new feature vectors. Then hypothesis testing methods apply:

1. Compare the R square coefficients before and after adding the polynomial terms.

2. Or see if the p-values for the polynomial terms are significant.

Corn Yield Data

Let's consider an example, modeling corn yield on amount of rainfall. Below is documentation for the data set:

response

```
yield: yearly corn yield in bushels per acre, in six Corn Belt
states  (Iowa, Illinois, Nebraska, Missouri, Indiana, and
Ohio).

rain: rainfall measurements in inches,  in  the  six  states.
from 1890 to 1927.

SOURCE:
 E.  G. Misner,
 "Studies  of  the  Relationship of Weather to the
 Production  and  Price  of  Farm  Products,  I.  Corn",
 mimeographed publication, Cornell University, March 1928.
```

(Based on the rainfall levels given, it is probably rainfall over the growing season, rather than annual rainfall.)

Let's assume the perspective of E. G. Misner back in 1928 at Cornell, doing this research. (Poor Dr. Misner didn't have R software, so we have an advantage.)

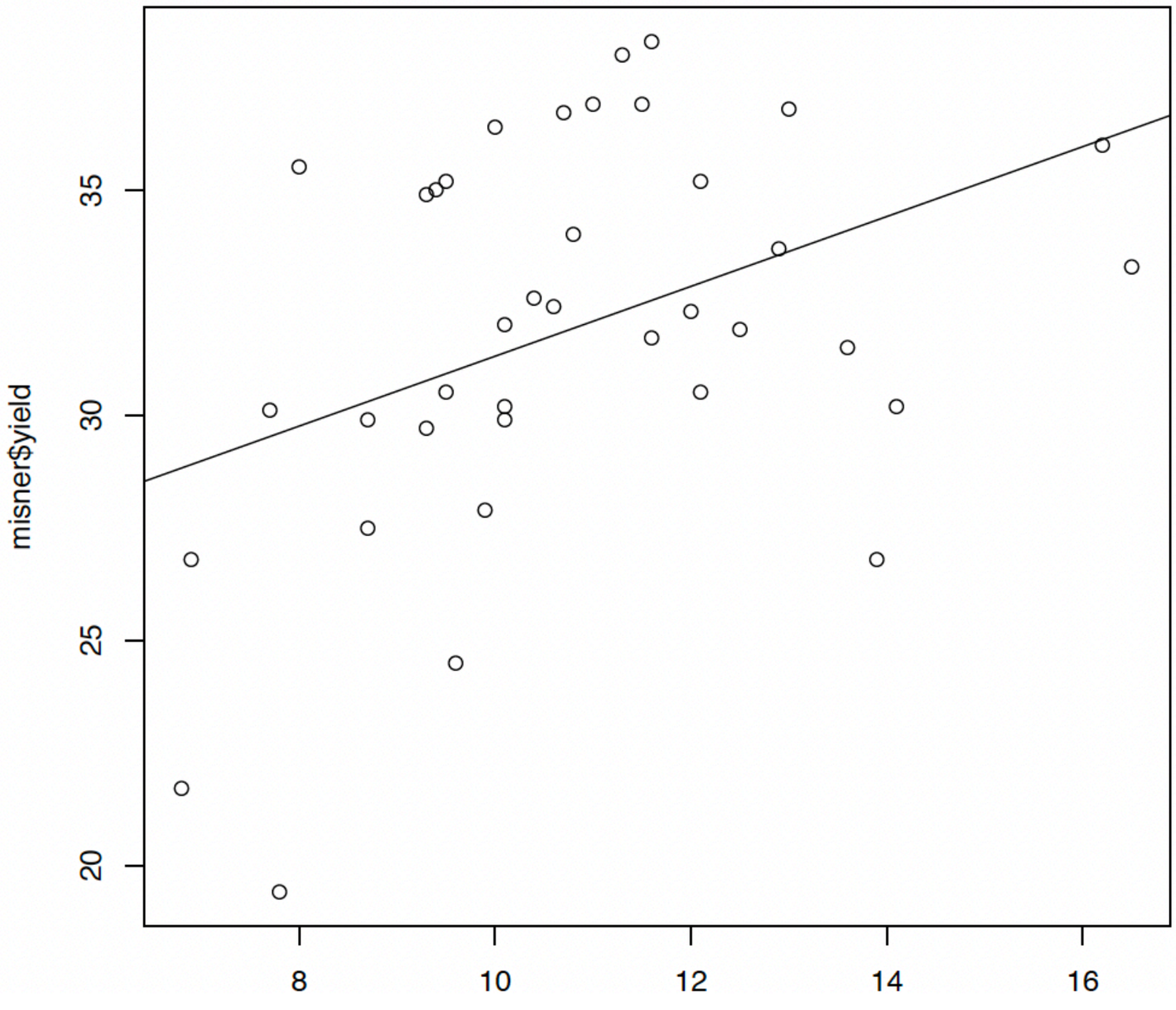
```
> misner <- read.table("misner.dat", header=TRUE)

> # First plot corn yield versus rain, with simple linear regression included

> plot(misner$rain, misner$yield)

> abline(lsfit(misner$rain, misner$yield))
```

Yield vs Rainfall



Simple Linear Regression

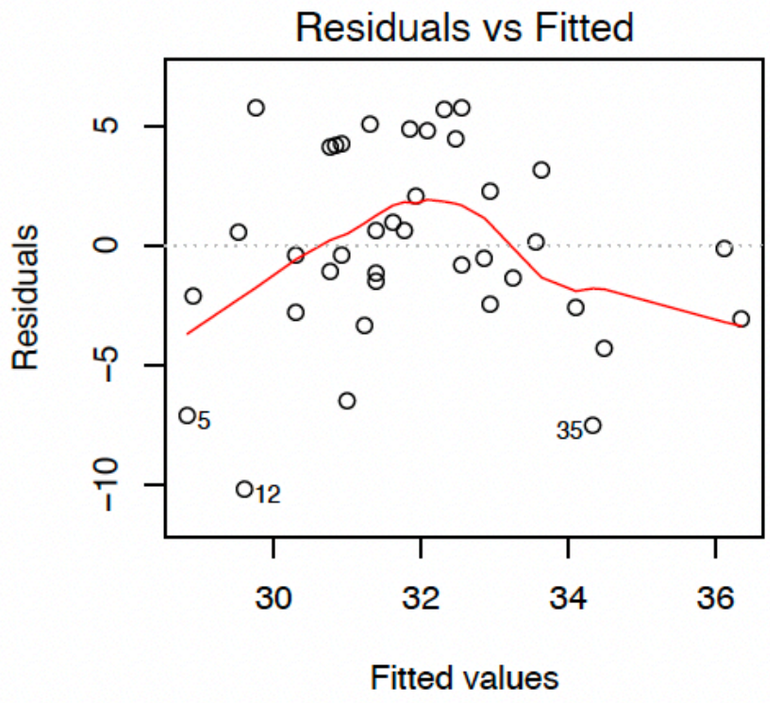
It looks like the benefit of rain diminishes after so many inches, producing a curvilinear relationship. Let's examine a simple linear regression and its diagnostic plots:

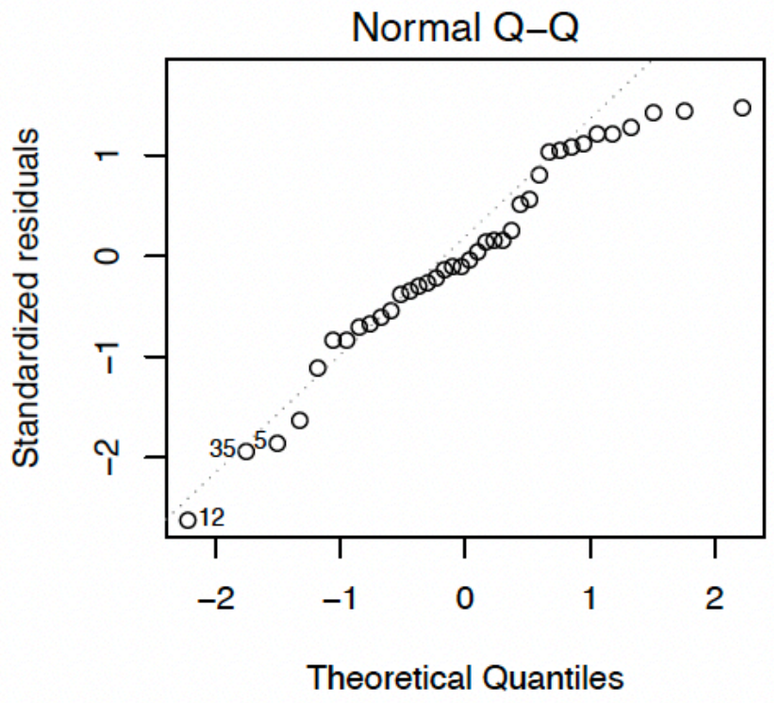
```
> model1 <- lm(yield ~ rain, data=misner)
> summary(model1)
...

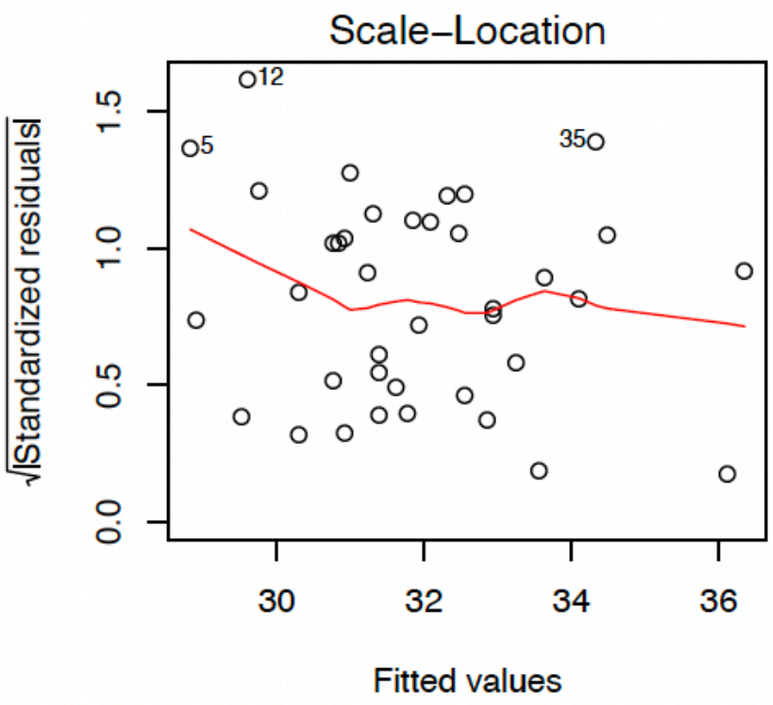
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  23.5521     3.2365   7.277 1.43e-08 ***
rain          0.7755     0.2939   2.639   0.0122 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

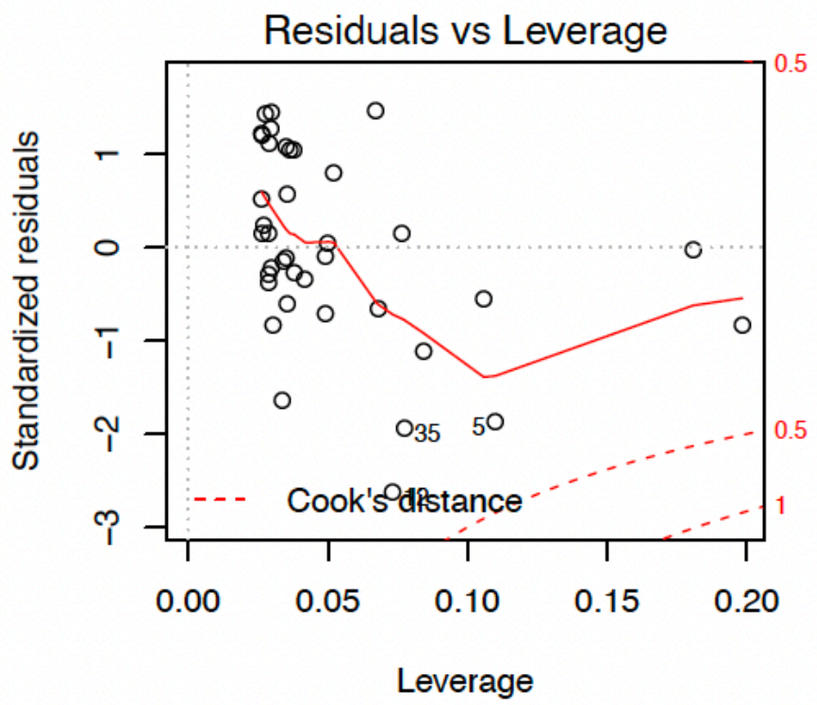
Residual standard error: 4.049 on 36 degrees of freedom
Multiple R-squared:  0.1621,Adjusted R-squared:  0.1388
F-statistic: 6.965 on 1 and 36 DF,  p-value: 0.01221

> par(mfrow=c(2,2))
> plot(model1)
```


Residuals vs Fitted
Residuals
Fitted values
5
12
35


Normal Q–Q
Standardized residuals
Theoretical Quantiles
35
5
12


Scale–Location
√|Standardized residuals|
Fitted values
12
5
35


Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
0.5
1
35
5
12

Quadratic Regression

The Residuals-vs-Fitted plot suggests an inverted U-shape. A quadratic term in the model might help ...

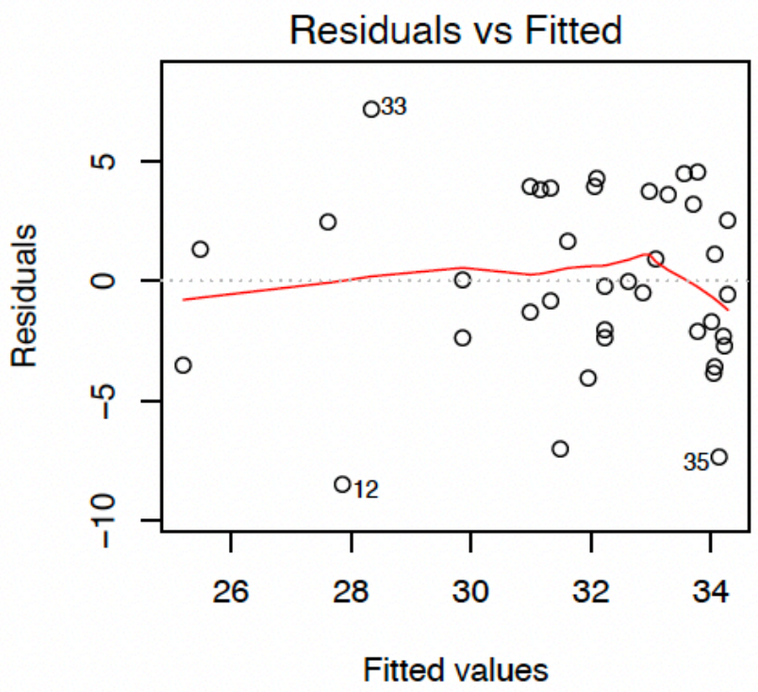
```
> model2 <- lm(yield ~ rain + I(rain^2), data=misner)
> summary(model2)
...

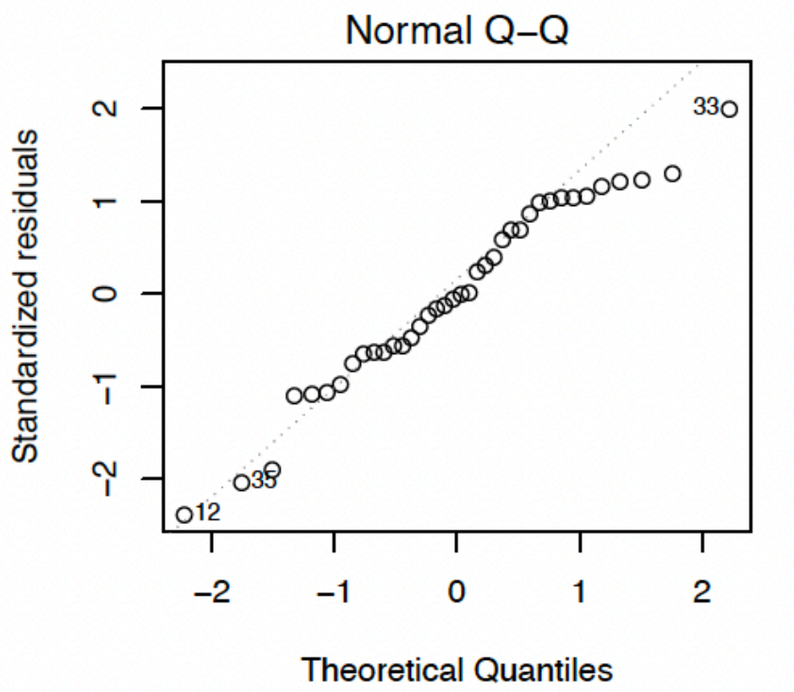
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.01467   11.44158  -0.438  0.66387
rain         6.00428    2.03895   2.945  0.00571 **
I(rain^2)   -0.22936    0.08864  -2.588  0.01397 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

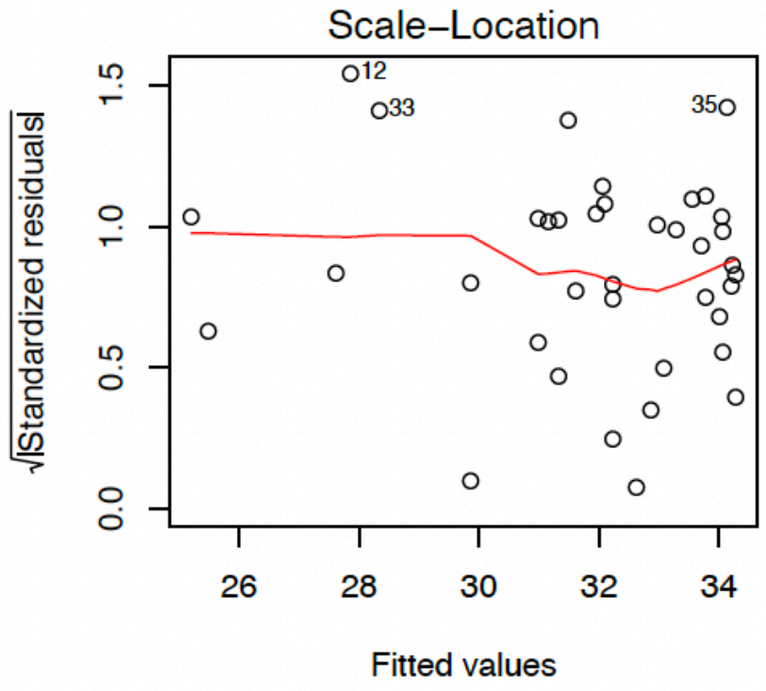
Residual standard error: 3.763 on 35 degrees of freedom
Multiple R-squared:  0.2967,Adjusted R-squared:  0.2565
F-statistic: 7.382 on 2 and 35 DF,  p-value: 0.002115

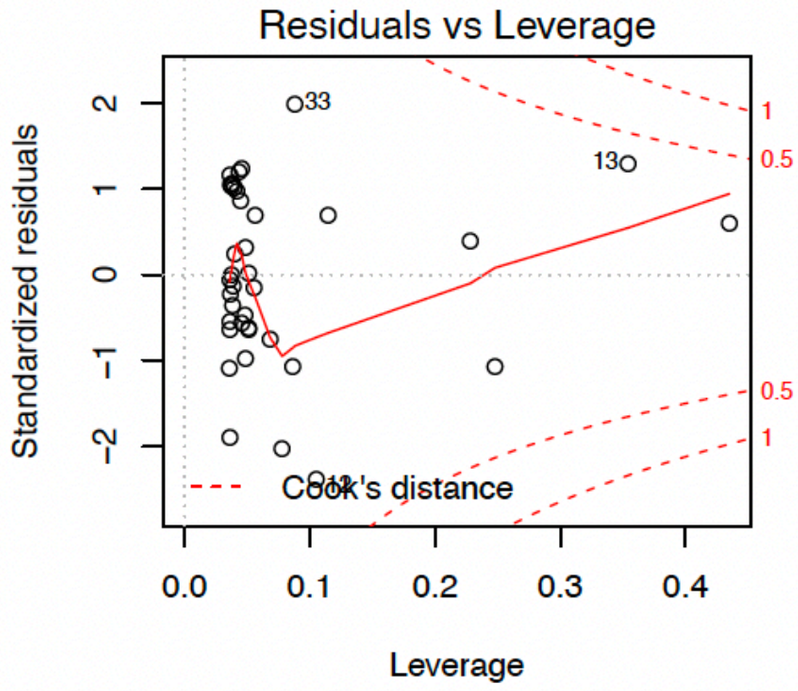
> par(mfrow=c(2,2))
> plot(model2)
```

I(...) isolate the contents in ()


Residuals vs Fitted
Residuals
Fitted values
33
12
35
26
28
30
32
34
5
0
−5
−10


Normal Q–Q
Standardized residuals
Theoretical Quantiles
33
35
12
−2
−1
0
1
2


Scale–Location
√|Standardized residuals|
Fitted values
12
33
35
26
28
30
32
34
0.0
0.5
1.0
1.5

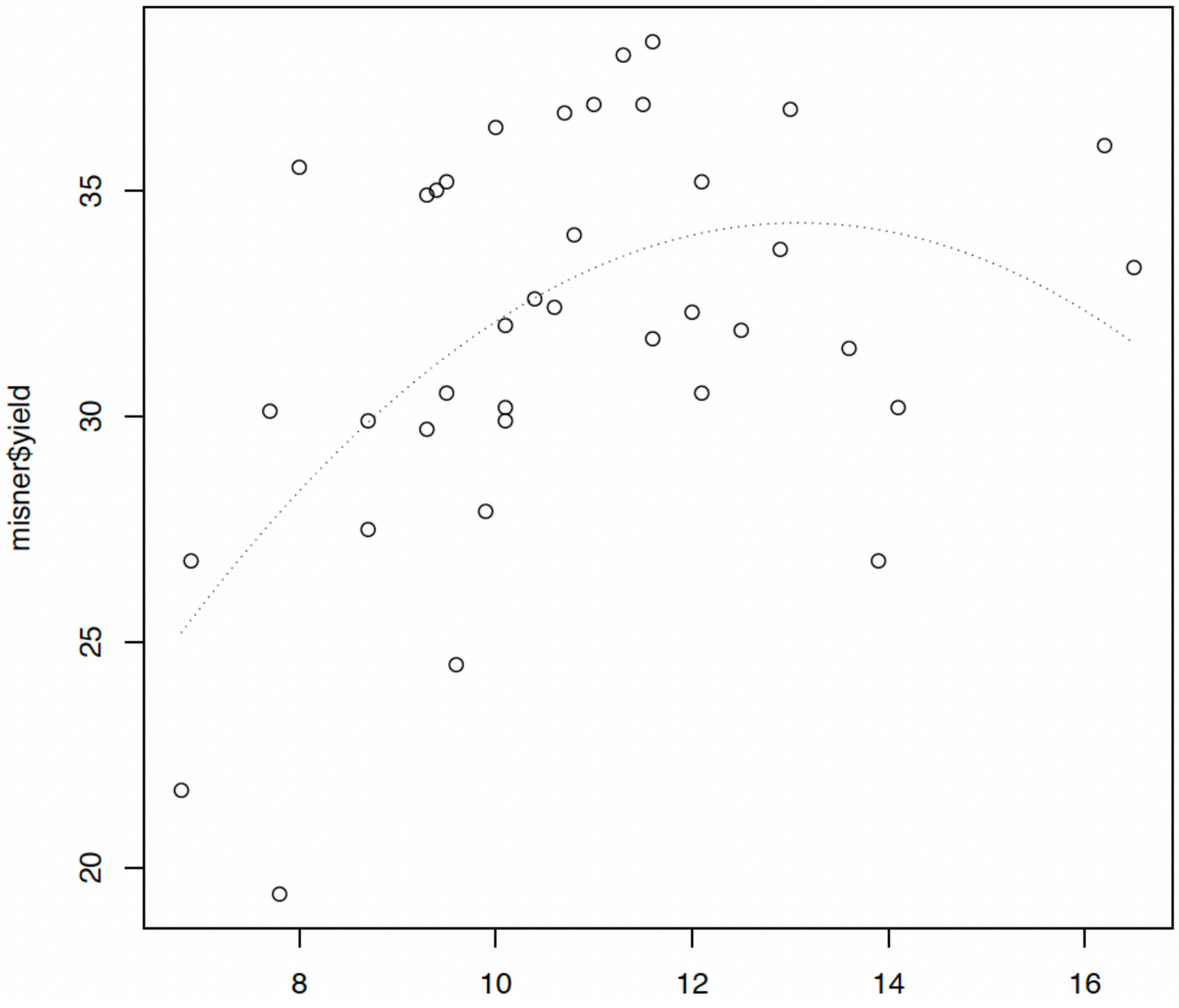

Residuals vs Leverage
Standardized residuals
Leverage
Cook's distance
33
13
12
1
0.5
0.5
1
0.0
0.1
0.2
0.3
0.4
−2
−1
0
1
2

Quadratic fit

We can plot the quadratic fit on the yield-versus-rain plot for a direct visual examination ...

```
> par(mfrow=c(1,1))
> plot(misner$rain, misner$yield)
> x.grid <- seq(min(misner$rain), max(misner$rain), len=100)
> lines(x.grid, predict(model2, data.frame(rain=x.grid)), lty=3)
```

Quadratic fit plot



R^2 improvement

- The R^2 for the simple linear regression was 0.1621, while the R^2 for the quadratic fit was 0.2967. So R^2 nearly doubles by adding the quadratic term.
- The p-value for testing the linear versus quadratic fit is the same as the p-value for the significance of the coefficient for the quadratic term: 0.01397.
- Dr. Misner would probably say that R^2 is quite impressive for the quadratic fit. However, he would note that corn yields have improved over time due to advances in irrigation, fertilizer, insect and weed control, etc. So we should consider adjusting for a time trend.

Time included as a regressor

```
> # Now also adjust for a time trend

> time <- 1890:1927

> model3 <- lm(yield ~ rain + I(rain^2) + time, data=misner)
> summary(model3)
...

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -263.30324   98.24094  -2.680  0.01126 *
rain           5.67038    1.88824   3.003  0.00499 **
I(rain^2)     -0.21550    0.08207  -2.626  0.01286 *
time           0.13634    0.05156   2.644  0.01229 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.477 on 34 degrees of freedom
Multiple R-squared:  0.4167,Adjusted R-squared:  0.3652
F-statistic: 8.095 on 3 and 34 DF,  p-value: 0.0003339
```

Quadratic term for time

Let's now fit a quadratic term for time, and see if it it needed.

```
> model4 <- lm(yield ~ rain + I(rain^2) + time + I(time^2), data=misner)
> summary(model4)
...

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.904e+04  1.914e+04  -1.517  0.13873
rain         6.189e+00  1.886e+00   3.282  0.00244 **
I(rain^2)   -2.389e-01  8.208e-02  -2.911  0.00642 **
time         3.029e+01  2.005e+01   1.510  0.14049
I(time^2)   -7.899e-03  5.254e-03  -1.503  0.14222
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.414 on 33 degrees of freedom
Multiple R-squared:  0.4541,Adjusted R-squared:  0.3879
F-statistic: 6.861 on 4 and 33 DF,  p-value: 0.0003908
```

The quadratic term increases R^2 by only about 0.04 and the p-value for `time^2` is 0.14222, so only a linear term for time.

Interaction between rain and time

Finally, let's consider if time and rainfall interact with one another.

Let Y=yield, X_1=rain, and X_2=time. We arrive at the model

$$Y = \beta_0 + \beta_1 X_1 + \beta_{11} X_1^2 + \beta_2 X_2 + \beta_{12} X_1 X_2 + \varepsilon$$

The parameter β_{12} measures the interaction between rain and time. Suppose we fix rainfall, X_1, and increase time by one year. Then, what is the expected change in Y?

$$\mathrm{E}(\Delta Y) = \beta_2 + \beta_{12} X_1$$

The interpretation is that yield changes with time, but the effect of time depends on rainfall.

An example of non-full-rank design

- Is the design matrix full rank, assuming distinct feature vectors across the observations?
- Consider the example where $x_{1i} = \frac{1}{x_{2i}}$, $i = 1, \ldots, n$.

```
> model5 <- lm(yield ~ rain + I(rain^2) + time + time*rain, data=misner)
> summary(model5)
...

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -1.909e+03  4.862e+02  -3.927 0.000414 ***
rain         1.588e+02  4.457e+01   3.564 0.001138 **
I(rain^2)   -1.862e-01  7.198e-02  -2.588 0.014257 *
time         1.001e+00  2.555e-01   3.919 0.000423 ***
rain:time   -8.064e-02  2.345e-02  -3.439 0.001599 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 3.028 on 33 degrees of freedom
Multiple R-squared:  0.5706,Adjusted R-squared:  0.5185
F-statistic: 10.96 on 4 and 33 DF,  p-value: 9.127e-06
```

Regression Splines

Disadvantage of polynomials: each data point affects the fit globally. Remedy: B-spline.
Cubic B-spline basis functions on interval (a, b) with pre-specified knots $t_1, \ldots, t_k$:

- Non-zero on interval defined by **five** successive knots and zero elsewhere $\Rightarrow$ local influence property
- Cubic polynomial fit to each four successive knots
- Smooth
- Integrates to one

Simulation Example

$$y = \sin^3\left(2\pi x^3\right) + \epsilon, \quad \epsilon \sim N(0, 0.1^2)$$

- Not a polynomial, not a cubic spline...
- But smooth and has many inflection points

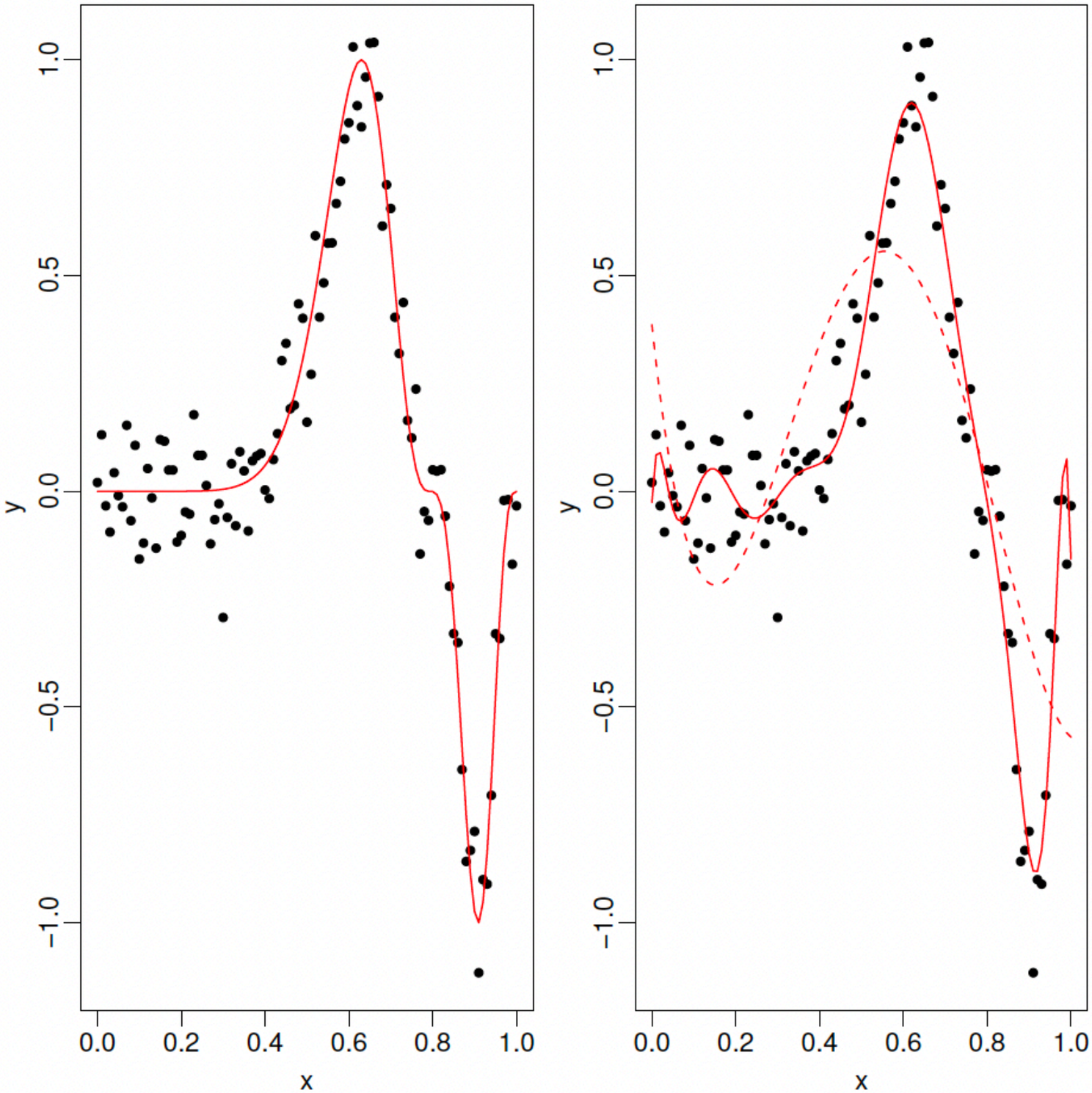
Simulation Example

```
## Data generation
> myf = function(x) sin(2*pi*x^3)^3
> x = seq(0, 1, by=0.01)
> y = myf(x) + 0.1*rnorm(101)
> matplot(x, cbind(y, myf(x)), type="pl")

## Polynomials
> g4 = lm(y ~ poly(x, 4))
> g12 = lm(y ~ poly(x, 12))
> matplot(x, cbind(y, g4$fit, g12$fit),  type="pll")
```

degree of polynomial

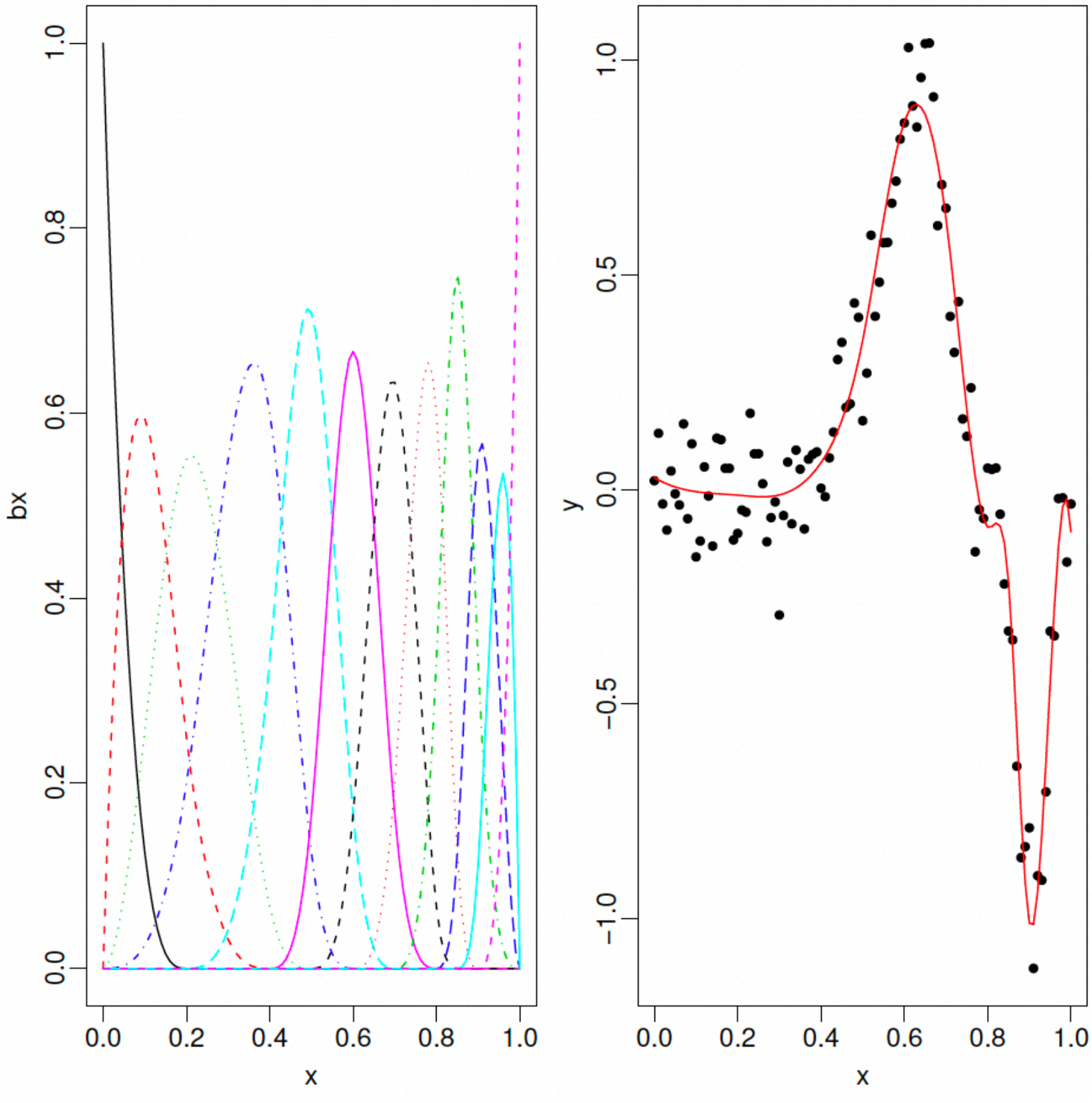
Polynomial results



```
## Regression splines
> library(splines)
> knots = c(0, 0, 0, 0, 0.2, 0.4, 0.5, 0.6,
    0.7, 0.8, 0.85, 0.9, 1, 1, 1, 1)
> bx = splineDesign(knots, x)
> gs = lm(y ~ bx)
> matplot(x, bx, type="l")
> matplot(x, cbind(y, gs$fit), type="pl")
```

creates a design matrix with spline functions as its columns

Spline results



Other Transformations

classification and regression tree

- Smoothing splines
- Generalized additive models
- CART, MARS, MART, neural networks

multivariate adaptive regression spline

Rule of thumb:

– for large data sets, complex models are better (with appropriate control of the number of parameters);

– for small data sets or high noise levels (e.g., social sciences), standard regression is more appropriate.

General conclusion

Transformations of X (including polynomials) can effectively treat the problem of curvature in the mean function, but not the problem of variance inhomogeneity (heteroscedasticity).

A transformation of Y *can* treat the problem of heteroscedasticity.

Transformation of both X and Y may be needed to treat both a problem of curvature and a problem of heteroscedasticity.

Thank you!

We gratefully acknowledge the previous course instructor Naveen Narisetty and various online sources for the images and slides presented.